

LECTURE 8: CHARACTERISTIC FUNCTIONS AND CENTRAL LIMIT THEOREM

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to elre@princeton.edu, thanks in advance!

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INTRODUCTION: DISTRIBUTIONAL LIMIT OF THE SUM OF INDEPENDENT RANDOM VARIABLES

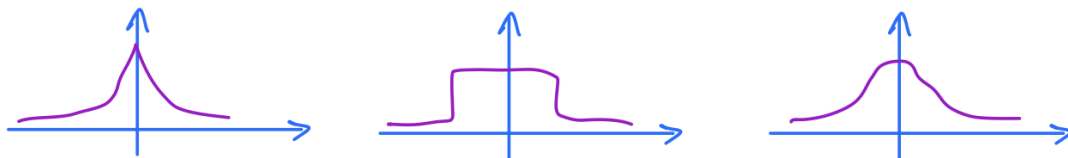
The law of large numbers shows that under very mild distributional assumptions, the averages of independent mean zero random variables converge to zero. This holds almost surely, and thus also in probability, in distribution (weakly), and also possibly in L^p .

It is interesting to get more delicate results on the convergence of independent sums, not only in the nontrivial case where the limiting law “shrinks” to a constant. Note that convergence to a constant is somewhat predetermined by the “scaling” (averaging). Indeed,

$$\text{for } \tilde{S}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \text{we have} \quad \text{Var}(\tilde{S}_n) = \frac{1}{n^2} \text{Var} \left(\sum_{i=1}^n X_i \right) = \frac{\text{Var}(X_1)}{n} \rightarrow 0,$$

as $n \rightarrow \infty$, which shows by averaging, we kept the expectation of the same order, but shrinks the “average spread” to zero.

This suggests that we should scale random variables in a different way for the nontriviality of the limiting distribution of the sum, keeping the variance of the same order. Then we can ask: what shape does the sum of independent random variables have with this scaling as $n \rightarrow \infty$?



The **Central Limit Theorem (CLT)** claims that the limit (appropriately scaled) is the Gaussian distribution $N(0,1)$ with mean zero and variance one, which has the famous bell-shaped curve.

Theorem 8.1 (CLT for i.i.d. sums). *Let $X_1, X_2, \dots$ be i.i.d. random variables with finite second moments. Define $\tilde{S}_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mathbb{E}[X_i])$. Then*

$$\frac{\tilde{S}_n}{\sqrt{\text{Var}[\tilde{S}_n]}} = \sum_{i=1}^n \frac{1}{\sqrt{n}} \left(\frac{X_i - \mathbb{E}[X_i]}{\sqrt{\text{Var}[X_i]}} \right) \xrightarrow{w} X,$$

where the law of X has standard normal distribution $N(0, 1)$.

This is a *universality* result: generic distributions of X_i lead to the behavior of $N(0, 1)$ in a properly taken limit.

Why is it that the normal distribution that arises? One piece of evidence is that the normal distribution is well-behaved:

Lemma 8.2 (Sums of Gaussians). *Let $X_i \sim N(\mu_i, \sigma_i^2)$ be mutually independent random variables, each having the normal distribution with mean μ_i and variance σ_i^2 . Then for all k , $S_k := \sum_{i=1}^k X_i$ is normally distributed with $S_k \sim N(\sum_{i=1}^k \mu_i, \sum_{i=1}^k \sigma_i^2)$.*

Proof. It is enough to prove the claim for a sum of two random variables (think this through). Furthermore, we can assume that the random variables are mean zero: indeed, $X \sim N(\mu_1, \sigma_1^2)$ if and only if $X - \mu_1 \sim N(0, \sigma_1^2)$ since

$$\mathbb{P}(X - \mu_1 \leq x) = \mathbb{P}(X \leq \mu_1 + x) = \frac{1}{\sqrt{2\pi}\sigma_1} \int_{-\infty}^{x+\mu_1} e^{-\frac{(t-\mu_1)^2}{2\sigma_1^2}} dt = \frac{1}{\sqrt{2\pi}\sigma_1} \int_{-\infty}^x e^{-\frac{t^2}{2\sigma_1^2}} dt \quad (1)$$

by the change of variables formula.

For independent $X_1 \sim N(0, \sigma_1^2)$ and $X_2 \sim N(0, \sigma_2^2)$, their sum has the density defined by the convolution of two Gaussian densities, namely,

$$f_{X_1+X_2}(x) = \frac{1}{\sqrt{2\pi}\sigma_1} \frac{1}{\sqrt{2\pi}\sigma_2} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{2\sigma_1^2}} e^{-\frac{y^2}{2\sigma_2^2}} dy = \frac{1}{\sqrt{2\pi}\sqrt{\sigma_1^2 + \sigma_2^2}} e^{-\frac{x^2}{2(\sigma_1^2 + \sigma_2^2)}} \cdot Z,$$

where

$$Z := \frac{1}{\sqrt{2\pi}\sigma} \int_{\mathbb{R}} e^{-\frac{1}{2\sigma^2} \left[-\sigma^2 \frac{x^2}{(\sigma_1^2 + \sigma_2^2)} + \sigma^2 \frac{(x-y)^2}{\sigma_1^2} + \sigma^2 \frac{y^2}{\sigma_2^2} \right]} dy \quad \text{for } \sigma := \frac{\sigma_1\sigma_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}.$$

A direct check with some algebra and regrouping of terms shows that the square bracket is a complete square of the type $(y - \dots)^2$, and therefore $Z = 1$ as an integral of a Gaussian density. $\square$

So, for Gaussian random variables, the CLT (Theorem 8.1) holds *exactly*, and not only in the limit but for all finite n . If there is universality in Theorem 8.1, it must be with the Gaussian weak limit. Note that taking limits differently can lead to different limiting (stable) laws.

CHARACTERISTIC FUNCTIONS

We have learned that for checking convergence in distributions, it is enough to check that $\mathbb{E}(g(X_n)) \rightarrow \mathbb{E}(g(X))$ for any bounded and continuous function g . We will prove and use an even more convenient criterion based on **characteristics functions**:

Definition 8.3. The characteristic function of a random variable X is the (complex-valued) function $\phi_X : \mathbb{R} \rightarrow \mathbb{C}$ defined by $\phi_X(t) = \mathbb{E}e^{itX}$.

Note that characteristic functions always exist (since $|e^{itX}| = 1$). For us, the crucial property is that *characteristic functions uniquely determine distributions*, and that their *pointwise convergence captures the convergence in distribution*:

Theorem 8.4 (Characteristic functions). *The following holds:*

- **[Uniqueness]** Random variables X and Y have the same CDF $F_X = F_Y$ if and only if they have the same characteristic functions $\phi_X = \phi_Y$.
- **[Lévy's continuity theorem]** Let $\{X_n\}_{n \geq 1}$ be a sequence of random variables such that the characteristic functions converge pointwise to a function $\phi : \mathbb{R} \rightarrow \mathbb{C}$ that is continuous at zero ($\phi_{X_n}(t) \rightarrow \phi(t)$ for all $t \in \mathbb{R}$). Then there exists a random variable X such that $\phi_X = \phi$ and $X_n \xrightarrow{w} X$.

We will prove this in the next section. Let us list other useful properties of characteristic functions that will be used for the proof of Theorem 8.1.

Proposition 8.5 (Properties of characteristic functions). *Let X be a random variable with characteristic function ϕ_X . The following properties hold:*

- (1) **Boundedness and normalization:** A simple check reveals that for all $t \in \mathbb{R}$ and $a, b \in \mathbb{R}$,

$$|\phi_X(t)| \leq 1, \quad \phi_X(0) = 1, \quad \phi_{aX+b}(t) = e^{itb} \phi_X(at).$$

- (2) **Uniform continuity:** For any $t, h \in \mathbb{R}$,

$$|\phi_X(t+h) - \phi_X(t)| = |\mathbb{E} e^{itX} (e^{ihX} - 1)| \leq \mathbb{E} |e^{ihX} - 1| \xrightarrow{h \rightarrow 0} 0$$

by the dominated convergence theorem. Note that the rate of convergence does not depend on the point t . This property is called uniform continuity.

- (3) **Characteristic function of a (independent) sum:** If X and Y are independent RVs then $\phi_{X+Y}(t) = \phi_X(t) \phi_Y(t)$ for any $t \in \mathbb{R}$. (This follows from the result about the expectation of products of independent random variables from Lecture 6.)
- (4) **Derivatives computation:** If the n th moment of X exists (i.e., $\mathbb{E}|X|^n < \infty$), then the n th derivative of its characteristic function exists and can be computed by a convenient formula:

$$\phi_X^{(n)}(t) = i^n \mathbb{E}[X^n e^{itX}].$$

Proof of (4). We will prove the claim by induction. For the base case, we clearly have $\phi_X(t) = i^0 \mathbb{E}[X^0 e^{itX}] = \mathbb{E}[e^{itX}]$. Next, assuming that $\phi_X^{(k)}(t) = i^k \mathbb{E}[X^k e^{itX}]$ for $k < n$, we want to prove that the claimed formula holds for $\phi_X^{(k+1)}(t)$. By definition,

$$\phi_X^{(k+1)}(t) = \lim_{h \rightarrow 0} \frac{\phi_X^{(k)}(t+h) - \phi_X^{(k)}(t)}{h} = \lim_{h \rightarrow 0} \mathbb{E} \left[\frac{i^k X^k e^{itX} (e^{ihX} - 1)}{h} \right].$$

We will use the fact that

$$\lim_{h \rightarrow 0} \frac{e^{ihX} - 1}{h} \rightarrow iX \quad \text{pointwise.}$$

Thus, for some constant C , we can bound $|i^k X^k e^{itX} \frac{(e^{ihX} - 1)}{h}| \leq C|X^k||X| = C|X^{k+1}|$, which is integrable by assumption. So, by the dominated convergence theorem,

$$\phi_X^{(k+1)}(t) = \mathbb{E} \left[i^k X^k e^{itX} \lim_{h \rightarrow 0} \frac{(e^{ihX} - 1)}{h} \right] = i^{k+1} \mathbb{E} [X^{k+1} e^{itX}],$$

which completes the proof. □

Example 8.6. (1) Rademacher RV: $X = \pm 1$ with probability 1/2 each.

$$\phi_X(t) = \frac{e^{it} + e^{-it}}{2} = \cos t.$$

(2) Uniform RV: $X \sim \text{Unif}([-1, 1])$.

$$\phi_X(t) = \int_{-1}^1 e^{itx} \frac{1}{2} dx = \frac{e^{it} - e^{-it}}{2it} = \frac{\sin t}{t}.$$

(3) Gaussian RV: $X \sim N(0, 1)$.

$$\phi_X(t) = e^{-t^2/2}.$$

To show this in a way that avoids computing complex integrals, consider the following trick (using Proposition 8.5):

$$\begin{aligned} \phi'_X(t) &= i\mathbb{E}[Xe^{itX}] = i\mathbb{E}[X(\cos(tX) + i\sin(tX))] \\ &= i\mathbb{E}[X \cos(tX)] - \mathbb{E}[X \sin(tX)]. \end{aligned}$$

Now, the density function of a Gaussian random variable and the cosine function are both even, so $i\mathbb{E}[X \cos(tX)] = 0$ (integrating an odd function over the real line). Next, note that if $f_X(x)$ is the Gaussian density function $f_X(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$, we have the following relation $-xf_X(x) = f'_X(x)$. So, by integration by parts,

$$\begin{aligned} \phi'_X(t) &= -\mathbb{E}[X \sin(tX)] = -\int_{\mathbb{R}} x \sin(tx) f_X(x) dx \\ &= \int_{\mathbb{R}} \sin(tx) f'_X(x) dx \\ &= \sin(tx) f_X(x) \Big|_{-\infty}^{\infty} - \int_{\mathbb{R}} t \cos(tx) f_X(x) dx \\ &= 0 - t\phi_X(t), \end{aligned}$$

since $f_X(x) \rightarrow 0$ when $x \rightarrow \pm\infty$ and $\phi_X(t) = \mathbb{E}[\cos(tX) + i\sin(tX)] = \mathbb{E}\cos(tX)$, again using that integrating an odd function $\sin(tx)f_X(x)$ over $\mathbb{R}$ is zero. The differential equation $\phi'_X(t) = -t\phi_X(t)$ that we have obtained implies that

$$\frac{d\phi}{\phi} = -t dt; \quad \ln(\phi) = -\frac{t^2}{2} + C; \quad \phi = Ce^{-t^2/2}.$$

Finally, $C = 1$ since $\phi_X(0) = 1$.

Corollary 8.7. (1) *Using part (1) of Proposition 8.5 and the fact that $X \sim N(\mu_1, \sigma_1^2)$ if and only if $X = \sigma(\tilde{X} + \mu)$ for some $\tilde{X} \sim N(0, 1)$ (which can be shown similarly to (1)), it is straightforward to check that*

$$\phi_X(t) = e^{it\mu - \frac{t^2\sigma^2}{2}} \quad \text{for } X \sim N(\mu, \sigma^2).$$

(2) *We can re-prove Lemma 8.2 using part (3) of Proposition 8.5 and the uniqueness part of Theorem 8.4. Indeed, for independent Gaussian random variables $X_1 \sim N(0, \sigma_1^2)$ and $X_2 \sim N(0, \sigma_2^2)$,*

$$\phi_{X_1+X_2}(t) = \phi_{X_1}(t)\phi_{X_2}(t) = e^{-\frac{t^2\sigma_1^2}{2} - \frac{t^2\sigma_2^2}{2}} = e^{-\frac{t^2}{2}(\sigma_1^2 + \sigma_2^2)} = \phi_Z(t),$$

where ϕ_Z is the characteristic function of $Z \sim N(0, \sigma_1^2 + \sigma_2^2)$.

PROOF OF THE CENTRAL LIMIT THEOREM 8.1

Let

$$\tilde{X}_i := \frac{X_i - \mathbb{E}X_i}{\sqrt{\text{Var}X_i}}.$$

Then $\mathbb{E}\tilde{X}_i = 0$ and $\text{Var}\tilde{X}_i^2 = 1$. We need to show that

$$Z_n := \sum_{i=1}^n \frac{1}{\sqrt{n}} \tilde{X}_i \xrightarrow{w} N(0, 1).$$

By the Lévy's continuity theorem (Theorem 8.4), it is enough to show that

$$\phi_{Z_n}(t) = \prod_{i=1}^n \phi_{\tilde{X}_i} \left(\frac{t}{\sqrt{n}} \right) = \prod_{i=1}^n \phi_{\tilde{X}_1} \left(\frac{t}{\sqrt{n}} \right) \xrightarrow{n \rightarrow \infty} e^{-\frac{t^2}{2}} \quad \text{for all } t \in \mathbb{R}.$$

Note that for all $t \in \mathbb{R}$, $\frac{t}{\sqrt{n}} \rightarrow 0$ as $n \rightarrow \infty$. Let us Taylor expand $\phi_{\tilde{X}_1}$ near zero. By Proposition 8.5, $\phi_{\tilde{X}_1}(0) = 1$, $\phi'_{\tilde{X}_1}(0) = i\mathbb{E}\tilde{X}_1 = 0$, and $\phi''_{\tilde{X}_1}(0) = i^2\mathbb{E}\tilde{X}_1^2 = -1$. Thus,

$$\phi_{\tilde{X}_1} \left(\frac{t}{\sqrt{n}} \right) = 1 - \frac{t^2}{2n} + \frac{t^2}{n} r \left(\frac{t^2}{n} \right), \quad \text{such that} \quad r(x) \xrightarrow{x \rightarrow 0} 0.$$

We claim that

$$\phi_{Z_n}(t) = \left[1 + \frac{-t^2/2 + t^2 r(t^2/n)}{n} \right]^n \rightarrow e^{-t^2/2} \quad \text{as } n \rightarrow \infty.$$

From the properties of characteristic functions (Proposition 8.5), it is enough to check the claimed limit to complete the proof since the right hand side is the characteristic function of $N(0, 1)$.

To prove the claimed convergence, we use the following fact: if $z_n \rightarrow z \in \mathbb{C}$, then $(1 + \frac{z_n}{n})^n \rightarrow e^z$. This fact can be shown using the following steps:

- If $|z| < c$, then it is easy to check that for large enough n , we also have $|z_n| < c$.
- If $z_1, z_2, \dots, z_n \in \mathbb{C}$ and $w_1, w_2, \dots, w_n \in \mathbb{C}$ are such that $|z_i|, |w_i| \leq \theta$ for every i , then

$$\left| \prod_{j=1}^n z_j - \prod_{j=1}^n w_j \right| \leq \theta^{n-1} \sum_{j=1}^n |z_j - w_j| \quad (\text{this is an exercise}).$$

- Using these properties, we deduce that

$$\left| \prod_{j=1}^n \left(1 + \frac{z_n}{n} \right) - \prod_{j=1}^n e^{z_n/n} \right| \leq e^{(c/n)(n-1)} \sum_{j=1}^n \left| 1 + \frac{z_n}{n} - e^{z_n/n} \right| \leq e^{(c/n)(n-1)} \cdot n \cdot \left| \frac{z_n}{n} \right|^2 \xrightarrow{n \rightarrow \infty} 0,$$

where the last inequality can be checked using the complex power series expansion for e^z .

- This implies $\left| (1 + \frac{z_n}{n})^n - e^{z_n} \right| \rightarrow 0$, and since $e^{z_n} \rightarrow e^z$ we get $(1 + \frac{z_n}{n})^n \rightarrow e^z$.

This completes the proof of the CLT for i.i.d. random variables. $\square$

PROOF OF THE CHARACTERISTIC FUNCTIONS THEOREM 8.4.

Part (a): Uniqueness. The statement follows from the following inversion formula for recovering the law of a random variable from its characteristic function:

Theorem 8.8 (Inversion formula). *Let X be a random variable with characteristic function ϕ_X and CDF F_X with $F_X(a) = \mathbb{P}(X \leq a)$. Then for any two continuity points $a < b$ of the CDF F_X ,*

$$\lim_{T \rightarrow \infty} \int_{-T}^T \frac{e^{-i\theta a} - e^{-i\theta b}}{2\pi i \theta} \phi_X(\theta) d\theta = F_X(b) - F_X(a).$$

Proof of Theorem 8.8. Let

$$\xi_{a,b}(\theta) := \frac{e^{-i\theta a} - e^{-i\theta b}}{2\pi i \theta}.$$

By Fubini's theorem (noting that the functions involved are bounded by constants),

$$\int_{-T}^T \xi_{a,b}(\theta) \phi_X(\theta) d\theta = \int_{-T}^T \xi_{a,b}(\theta) \int_{\mathbb{R}} e^{i\theta x} d\mathcal{P}_X(x) d\theta = \int_{\mathbb{R}} \int_{-T}^T \xi_{a,b}(\theta) e^{i\theta x} d\theta d\mathcal{P}_X(x).$$

Consider the inner integral:

$$\int_{-T}^T \xi_{a,b}(\theta) e^{i\theta x} d\theta = \int_{-T}^T \frac{e^{-i\theta a} - e^{-i\theta b}}{2\pi i \theta} e^{i\theta x} d\theta = R(x-a, T) - R(x-b, T),$$

where for $u, v \in \mathbb{R}$,

$$R(u, T) - R(v, T) = \int_{-T}^T \frac{e^{i\theta u} - e^{i\theta v}}{2\pi i \theta} d\theta = \int_{-T}^T \frac{i \sin(\theta u) - i \sin(\theta v)}{2\pi i \theta} d\theta.$$

using the fact that the integral of an odd function on $[-T, T]$ is zero. Observe that by doing a change of variable and using the fact that $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$,

$$\begin{aligned} \int_{-T}^T \frac{i \sin(\theta u)}{2\pi i \theta} d\theta &= \frac{\text{sign}(u)}{\pi} \int_0^T \frac{\sin(\theta|u|)}{\theta} d\theta \\ &= \frac{\text{sign}(u)}{\pi} \int_0^{T|u|} \frac{\sin(x)}{x} dx \xrightarrow{T \rightarrow \infty} \frac{1}{2} \text{sign}(u), \end{aligned}$$

Hence,

$$\lim_{T \rightarrow \infty} (R(x-a, T) - R(x-b, T)) = \frac{1}{2} (\text{sign}(x-a) - \text{sign}(x-b)) = \begin{cases} 1 & \text{on } (a, b), \\ \frac{1}{2} & \text{at } a \text{ and } b, \\ 0 & \text{outside } [a, b]. \end{cases}$$

Finally, by the dominated convergence theorem,

$$\lim_{T \rightarrow \infty} \int_{-T}^T \xi_{a,b}(\theta) \phi_X(\theta) d\theta = \int_{\mathbb{R}} \mathbf{1}_{(a,b)}(x) d\mathcal{P}_X(x) = \mathbb{P}(X \in (a, b)).$$

In the last step we used the assumption that a and b are both continuity points. This completes the proof. $\square$

Remark 8.9. If we denote $F_X(a-) := \mathbb{P}(X < a)$, then for any $a < b$ (not necessarily continuity points), we actually have

$$\lim_{T \rightarrow \infty} \int_{-T}^T \xi_{a,b}(\theta) \phi_X(\theta) d\theta = \frac{1}{2}(F_X(b) + F_X(b-)) - \frac{1}{2}(F_X(a) + F_X(a-)).$$

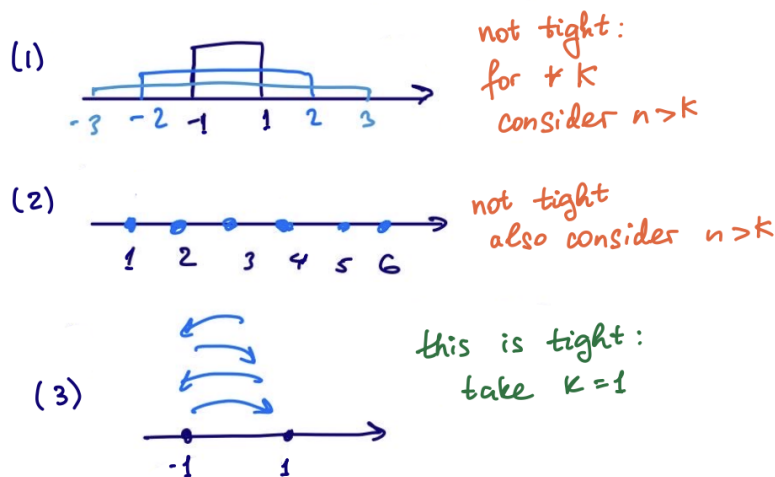
Moreover, if $\phi_X(\theta)$ is Lebesgue-integrable on $\mathbb{R}$, then $f_X(x) := \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\theta x} \phi_X(\theta) d\theta$ is the continuous and bounded density of X . See, e.g., [Dembo, Theorem 3.3.13] for the details.

Proof of Theorem 8.4 (uniqueness). If $F_X = F_Y$, then X and Y have the same law and hence the integrals defining their respective characteristic functions coincide. Conversely, suppose that $\phi_X = \phi_Y$ pointwise. Then Theorem 8.8 (with $a \rightarrow -\infty$) implies that $F_X(b) = F_Y(b)$ for every point of continuity b of F_X and F_Y . Recall that a CDF can have only countably many points of discontinuity and is right-continuous everywhere. Hence, for any $t \in \mathbb{R}$ we can approximate $t = \lim b_n$ by a decreasing limit of points of continuity of both F_X and F_Y , and we conclude that $F_X(t) = F_Y(t)$ for all $t \in \mathbb{R}$. $\square$

Part (b): Convergence. We need preliminary developments to prove the second part of Theorem 8.4 (Lévy's continuity theorem). It follows from the fact that pointwise convergence of characte *tightness property* of a sequence of measures

Definition 8.10. A sequence of probability measures $\{\mathbb{P}_n\}_{n \geq 1}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is called **tight** if for any $\varepsilon > 0$, there exists $K < \infty$ so that $\mathbb{P}_n([-K, K]) \geq 1 - \varepsilon$ for all n .

Informally, the tightness property can be understood as saying that “the mass does not escape to infinity”. Consider the following examples:



- (1) $\text{Unif}[-n, n]$ for $n = 1, 2, \dots$ is not tight since the mass of any compact interval goes to zero.
- (2) δ_n for $n = 1, 2, \dots$ is not tight since the point mass “escapes to infinity”
- (3) $\delta_{(-1)^n}$ for $n = 1, 2, \dots$ is tight: e.g., we can simply take $K = 1$.

Do the random variables converge weakly in any of these examples? No. However, observe that in the last case, we can extract a converging subsequence (e.g., $\delta_{(-1)^{n_k}}$ for $n_k = 2, 4, 6, \dots$). This is guaranteed by the following theorem:

Theorem 8.11 (Helly's selection theorem). *If $\{X_n\}_{n \geq 1}$ is sequence of random variables whose laws are tight,¹ then it has a subsequence $\{n_k\}_{k=1}^\infty$ that converges weakly.*

Proof. Let F_n denote the CDF of each X_n . By compactness, we can choose a subsequence $\{n_k\}_{k \geq 1}$ such that

$$F_*(q) := \lim_{k \rightarrow \infty} F_{n_k}(q)$$

exists for all rational $q \in \mathbb{Q}$.² We will extend F_* to the function $F : \mathbb{R} \rightarrow [0, 1]$ by defining

$$F(x) := \inf_{q \in \mathbb{Q}, q \geq x} F_*(q) \quad \forall x \in \mathbb{R}.$$

We claim that F is a CDF. Specifically, one can check that it is right-continuous, nondecreasing and has limits 0 and 1 when x tends to negative and positive infinity respectively (recall Lecture 3):

- By construction, F is non-decreasing.
- For any $x \in \mathbb{R}$, there exists a rational sequence $q_1 > q_2 > \dots$ decreasing to x such that $F(x) = \lim_{n \rightarrow \infty} F_*(q_n)$. Given any sequence $x_1 > x_2 > \dots$ of real numbers decreasing to x , for every k , we have $x_k < q_\ell$ for some $\ell = \ell(k)$. Since $F(x_k) \leq F_*(q_\ell)$,

$$\limsup_{k \rightarrow \infty} F(x_k) \geq F(x) = \lim_{n \rightarrow \infty} F_*(q_n) \geq \limsup_{k \rightarrow \infty} F(x_k),$$

which shows that F is right-continuous.

- By tightness, for any ε , there exists a rational number K such that $1 - F_n(K) = \mathbb{P}(X_n > K) \leq \mathbb{P}(|X_n| > K) \leq \varepsilon$ for all n , or rearranging, $F_n(K) \geq 1 - \varepsilon$. Thus,

$$F(K) = F_*(K) = \lim_{k \rightarrow \infty} F_{n_k}(K) \geq 1 - \varepsilon.$$

This shows that $\lim_{x \rightarrow \infty} F(x) = 1$, and the proof for $\lim_{x \rightarrow -\infty} F(x) = 0$ is similar.

Therefore, F is a CDF, which implies that there exists a random variable X whose CDF coincides with F .

To conclude, we claim that $\{X_{n_k}\}_{k \geq 1}$ converges weakly to X . This follows from the fact that F is almost everywhere continuous and that rational points are tight on $\mathbb{R}$ as follows:

- Let x be a continuity point of F . For any rational $q > x$, $F_{n_k}(x) < F_{n_k}(q)$ for all k , so

$$\limsup_{k \rightarrow \infty} F_{n_k}(x) \leq \lim_{k \rightarrow \infty} F_{n_k}(q) = F_*(q).$$

Since this holds for all rational $q > x$, we deduce that $\limsup_{k \rightarrow \infty} F_{n_k}(x) \leq F(x)$.

Next, take any $y < x$ and rational $q \in (y, x)$. Then

$$\liminf_{k \rightarrow \infty} F_{n_k}(x) \geq \lim_{k \rightarrow \infty} F_{n_k}(q) = F_*(q).$$

Since this holds for all rational $q \in (y, x)$,

$$\liminf_{k \rightarrow \infty} F_{n_k}(x) \geq F(y).$$

Since this holds for all $y < x$ and x is a continuity point of F , we deduce that $\liminf_{k \rightarrow \infty} F_{n_k}(x) \geq F(x)$. We conclude that $\lim_{k \rightarrow \infty} F_{n_k}(x) = F(x)$, which completes the proof.

¹That is, for all $\varepsilon > 0$, there exists $K < \infty$ such that $\mathbb{P}(|X_n| \leq K) > 1 - \varepsilon$ for all n

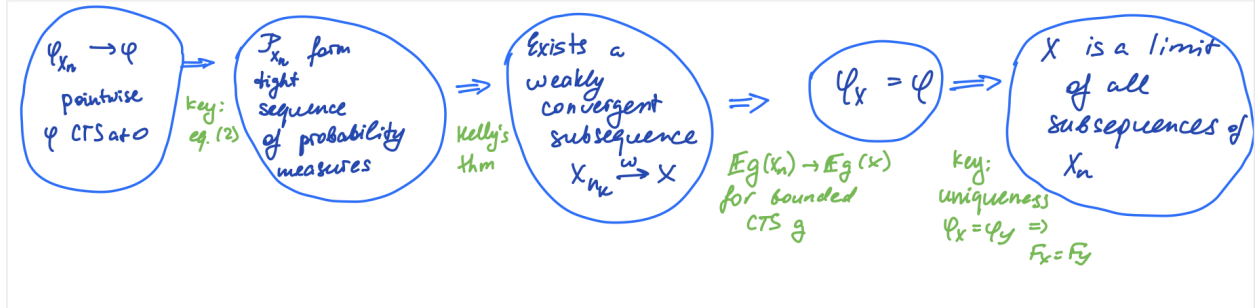
²compactness argument: for every fixed rational $q \in \mathbb{Q}$, sequence of numbers $\{F_n(q)\}_{n \geq 1} \subseteq [0, 1]$ is bounded sequence, so by Bolzano-Weierstrass theorem we can extract subsequence; extract subsequence for all countably many rational $\mathbb{Q}$ by diagonal argument

□

Exercise. (a) Show that if a sequence of random variables $\{X_n\}_{n \geq 1}$ converges weakly, then it must be tight. (b) Show that if for some $a > 0$ we have $\sup_n \mathbb{E}|X_n|^a < \infty$ then the system is tight. *Hint: use Markov's inequality.*

Therefore, a way to prove weak convergence of some sequence is through the following:

- (1) Prove that the sequence is tight
- (2) Take a weakly convergent subsequence (by Helly's theorem) and its limit X , check that $\phi_X = \phi$ as claimed
- (3) Show uniqueness of the limit



Now, we are going to use this strategy to prove the second part of Theorem 8.4.

Proof of Theorem 8.4 (Lévy's continuity theorem). **(1) Tightness** follows from the pointwise convergence of characteristic functions through the following computations. First, we bound the tails of the distribution by an integral involving the characteristic function. Namely, for any random variable X , note that we have

$$\int_{-1/K}^{1/K} [1 - \phi_X(t)] dt = \frac{2}{K} - \mathbb{E} \int_{-1/K}^{1/K} e^{itX} dt = \frac{2}{K} - \mathbb{E} \frac{e^{iX/K} - e^{-iX/K}}{iX} = \frac{2}{K} \left(1 - \mathbb{E} \frac{\sin(X/K)}{X/K} \right).$$

Then, using the fact that $1 - \frac{\sin t}{t} \geq 0$ for any $t \in \mathbb{R}$,

$$K \int_{-1/K}^{1/K} [1 - \phi_X(t)] dt \geq 2\mathbb{E} \left[\left(1 - \frac{\sin(X/K)}{X/K} \right) \mathbb{1}_{\{|X/K| > 2\}} \right] \geq 2\mathbb{E} \left[\left(1 - \frac{1}{|X|/K} \right) \mathbb{1}_{\{|X|/K > 2\}} \right],$$

since the function is even and $\sin t \leq 1$. The expression can be further lower bounded by $2\frac{1}{2}\mathbb{E}(\mathbb{1}_{\{|X|/K > 2\}}) = \mathbb{P}(|X| > 2K)$. Summing up, we have shown that

$$\mathbb{P}(|X| > 2K) \leq K \int_{-1/K}^{1/K} [1 - \phi_X(t)] dt. \quad (2)$$

Since $\phi(0) = 1$ (as a limit of $\phi_n(0) = 1$ for every n) and by continuity of ϕ at 0, we deduce that for all $\varepsilon > 0$, there exists $\delta > 0$ such that $|1 - \phi(t)| < \varepsilon$ for all $|t| < \delta$. So, for $K > 1/\delta$,

$$K \int_{-1/K}^{1/K} [1 - \phi(t)] dt \leq 2\varepsilon.$$

By the dominated convergence theorem,

$$K \int_{-1/K}^{1/K} [1 - \phi_{X_n}(t)] dt \rightarrow K \int_{-1/K}^{1/K} [1 - \phi(t)] dt$$

as $n \rightarrow \infty$, which implies that the former integral is at most 3ε for n large enough. So, using (2) with the random variable X_n , we conclude that $\mathbb{P}(|X_n| > 2K) \leq 3\varepsilon$ for all n large enough.

To get the statement for all n , we may simply increase K to ensure that the probability is at most 3ε for each of the finitely many members of the sequence before this point.

(2) The weak subsequence limit X and its characteristic function $\phi_X = \phi$. By tightness, Theorem 8.11 implies that there is a subsequence $X_{n_k} \xrightarrow{w} X$, and so $\phi_{n_k} \rightarrow \phi_X$ pointwise (since the characteristic function is a bounded and continuous function of a random variable). But since $\phi_{n_k} \rightarrow \phi$ pointwise, we must have $\phi_X = \phi$ pointwise. Thus, we can define X to be the random variable corresponding to the characteristic function ϕ .

(3) Uniqueness of the limit. Suppose that $X_n \not\xrightarrow{w} X$. Then there exists a subsequence $m_k \xrightarrow{k \rightarrow \infty} \infty$ so that for a continuous bounded function g and all indices k large enough

$$|\mathbb{E}g(X_{m_k}) - \mathbb{E}g(X)| > \varepsilon. \quad (3)$$

From the subsequence $\{m_k\}$ we can select a further subsequence that converges in distribution to some X' (by tightness and Theorem 8.11). By (3), $X \neq X'$ in distribution. However, by the same argument used in part (2), we must have $\phi_{X'} = \phi = \phi_X$. This contradicts the first part of the theorem, which states that the characteristic function uniquely identifies the law of a random variable. Hence, we conclude that $X_n \rightarrow X$, which completes the proof. $\square$

Remark 8.12. Why does continuity at zero matter? Consider the following example: let $Y \sim N(0, 1)$ and $X_n = nY$ for $n \geq 1$. Then as $n \rightarrow \infty$,

$$\phi_{X_n}(t) = \phi_{nY}(t) = \phi_Y(nt) = e^{-n^2 t^2 / 2} \rightarrow \begin{cases} 0 & \text{if } t \neq 0, \\ 1 & \text{if } t = 0. \end{cases}$$

The limit is discontinuous at 0. Observe that for all $t \in \mathbb{R}$,

$$F_{X_n}(t) = \mathbb{P}(Y \leq \frac{t}{n}) \rightarrow \mathbb{P}(Y \leq 0) = 0.5,$$

so the limit is not a CDF, so X_n does not converge in distribution.